

3 Special Test: Cervical Spine: Test for Vertigo & Dizziness tests: Vertebral Artery Test Dizziness Test etc. Test for Thoracic Outlet Syndrome: Adson's test, Roos stress Test etc. Test for Neurological symptoms: Foraminal Compression Test, Distraction Test, Shoulder Depression Test etc. PFDP 3 Cervical Special Tests .pdf **CERVICAL SPECIAL TESTS --- 1 Spurlings Test /Foraminal Compression Test/ PURPOSE** To assess cervical nerve root compression /cervical radiculopathy/ **PROCEDURE** • Patient sits comfortably • Extend laterally flex and rotate the head towards the affected side • Apply gentle downward compression on the head **POSITIVE SIGN** Radiating pain numbness or tingling into the upper limb **INDICATES** • Cervical nerve root compression • Cervical radiculopathy --- **2 Cervical Distraction Test PURPOSE** To assess cervical nerve root compression **PROCEDURE** • Patient lies supine • Therapist gently grasps the head and applies an upward traction force **POSITIVE SIGN** Relief of neck or arm pain **INDICATES** • Cervical radiculopathy • Foraminal compression --- **3 Upper Limb Tension Test /ULTT/ PURPOSE** To assess neural tissue mobility and detect peripheral nerve or cervical nerve root involvement --- **ULTT 1 /MEDIAN NERVE BIAS/ Procedure** • Depress the shoulder • Abduct the shoulder to 110 • Supinate the forearm • Extend the wrist and fingers • Externally rotate the shoulder • Slowly extend the elbow • Finally side flex the neck away if required **Positive Sign** • Pain tingling or numbness along the median nerve distribution • Difference in symptoms compared to the opposite side **Indicates** • Median nerve tension • Cervical radiculopathy --- **ULTT 2A /MEDIAN NERVE BIAS/ Procedure** • Shoulder depression • Elbow extension • External rotation of shoulder • Forearm supination • Wrist and finger extension **Positive Sign** Reproduction of median nerve symptoms **Indicates** Median nerve irritation --- **ULTT 2B /RADIAL NERVE BIAS/ Procedure** • Shoulder depression • Internal rotation of shoulder • Elbow extension • Forearm pronation • Wrist finger and thumb flexion with ulnar deviation **Positive Sign** Pain or tingling along the dorsolateral forearm and hand **Indicates** Radial nerve tension --- **ULTT 3 /ULNAR NERVE BIAS/ Procedure** • Shoulder depression • Shoulder abduction and external rotation • Elbow flexion • Forearm pronation • Wrist extension with radial deviation • Finger extension **Positive Sign** Pain or tingling along the medial forearm and little finger **Indicates** Ulnar nerve tension --- **4 Vertebral Artery Test PURPOSE** To assess vertebrobasilar insufficiency before cervical mobilization or manipulation **PROCEDURE** • Patient lies supine • Therapist extends rotates and laterally flexes the neck • Hold for about 30 seconds while observing symptoms **POSITIVE SIGN** • Dizziness • Vertigo • Nystagmus • Blurred vision • Nausea **INDICATES** Vertebrobasilar insufficiency **Precaution** Stop the test immediately if symptoms appear --- **SPECIAL EXAMINATION TESTS --- 1 DIZZINESS TEST PURPOSE** To assess for cervical vascular compromise/vertebrobasilar insufficiency that may produce dizziness during cervical movement **PROCEDURE** • Patient is positioned sitting • Examiner stabilizes the patients head • Slowly place the cervical spine into extension and rotation • Observe the patient for symptoms **POSITIVE SIGN** • Dizziness • Vertigo • Nausea • Visual disturbance • Nystagmus **INDICATES** Possible vertebrobasilar insufficiency and requires further medical evaluation --- **2 SHOULDER DEPRESSION TEST PURPOSE** To assess nerve root/nerve tissue irritation or shortening of neural structures in the cervical and upper limb region **PROCEDURE** • Patient is sitting • Examiner stabilizes/depresses the shoulder on the tested side • Passively side flex the patients neck away from the tested side • The position places tension on the cervical nerve structures **POSITIVE SIGN** • Reproduction of the patients radiating pain/paresthesia into the upper limb **INDICATES** Possible cervical nerve root or brachial plexus involvement --- **3 PINCH GRIP TEST PURPOSE** To assess the function of the anterior interosseous nerve /AIN/ and flexor pollicis longus/index FDP function **PROCEDURE** • Ask the patient to make a pinch/OK sign by touching the tip of the thumb to the tip of the index finger • Observe the shape of the circle formed **POSITIVE SIGN** • Patient cannot make a proper round OK circle • Instead the pinch becomes flattened/pulp to pulp **INDICATES** Anterior interosseous nerve palsy/dysfunction --- **4 CARPAL COMPRESSION TEST PURPOSE** To assess for carpal tunnel syndrome due to median nerve compression **PROCEDURE** • Patients wrist is in a neutral position • Examiner applies direct pressure over the carpal tunnel with the thumbs/fingers • Maintain pressure for approximately 30 seconds **POSITIVE SIGN** • Reproduction of • Pain • Tingling • Numbness • Paresthesia in the median nerve distribution **INDICATES** Carpal Tunnel Syndrome / median nerve compression --- **5 HAND ELEVATION TEST PURPOSE** To assess for carpal tunnel syndrome/median nerve compression **PROCEDURE** • Patient raises both hands above the head • Maintain the elevated position for approximately 1 2 minutes • Ask the patient about symptoms **POSITIVE SIGN** • Reproduction of pain numbness or tingling in the median nerve distribution **INDICATES** Possible Carpal Tunnel Syndrome --- **6 FROMENTS SIGN PURPOSE** To assess the ulnar nerve particularly the function of adductor pollicis **PROCEDURE** • Ask the patient to hold a piece of paper between the thumb and index finger • Examiner attempts to pull the paper away **POSITIVE SIGN** Patient compensates by • Flexing the IP joint of the thumb using flexor pollicis longus **INDICATES** Ulnar nerve palsy/weakness of adductor pollicis

4 Special Test: Shoulder Joint: Test for Shoulder Instability: Apprehension, Jerk, Drawer, Sulcus sign etc. Test for Impingement: Hawkins-Kennedy Test Neer's Impingement Test, Painful Arc Test, Empty Can Test, Drop Arm Test etc. Test for Labral Lesion: Active Compression test of O'Brien, KIM test etc. Test for Scapular Dyskinesia: Scapular Dyskinesia test,

Scapular Load Test etc. Test for Muscle Pathology: Yergason's, Speed's, Drop-Arm, Empty can, Lift-Off Sign, Test for Supraspinatus, Infraspinatus, Horn blower Sign etc. PFDP 4 Shoulder Special Tests.pdf --- **I INSTABILITY TESTS** ---

1 APPREHENSION TEST Purpose To assess anterior shoulder instability Procedure Abduct the shoulder to 90 flex the elbow to 90 and slowly externally rotate the shoulder Positive Sign Patient becomes apprehensive or fears shoulder dislocation Indicates Anterior glenohumeral instability---

2 JERK TEST Purpose To assess posterior shoulder instability and posterior labral lesions Procedure Shoulder abducted to 90 and internally rotated Apply axial compression through the elbow while horizontally adducting the arm Positive Sign Painful clunk or jerk Indicates Posterior instability or posterior labral tear---

3 DRAWER TEST Purpose To assess anterior and posterior glenohumeral instability Procedure Stabilize the scapula and translate the humeral head anteriorly and posteriorly Positive Sign Excessive humeral head movement compared to the opposite side Indicates Shoulder instability---

4 SULCUS SIGN Purpose To assess inferior glenohumeral instability Procedure Apply downward traction to the relaxed arm Positive Sign Visible sulcus below the acromion Indicates Inferior or multidirectional shoulder instability---

II IMPINGEMENT TESTS --- **1 HAWKINS KENNEDY TEST** Purpose Assess subacromial impingement Procedure Flex shoulder and elbow to 90 and internally rotate the shoulder Positive Sign Pain Indicates Subacromial impingement---

2 NEERS IMPINGEMENT TEST Purpose Assess subacromial impingement Procedure Stabilize the scapula and passively flex the internally rotated arm Positive Sign Pain Indicates Subacromial impingement---

3 PAINFUL ARC TEST Purpose Assess shoulder impingement Procedure Ask the patient to actively abduct the shoulder Positive Sign Pain between 60 120 of abduction Indicates Subacromial impingement---

4 EMPTY CAN JOBS TEST Purpose Assess supraspinatus tendon Procedure Shoulder abducted to 90 in the scapular plane internally rotated thumb down resist downward pressure Positive Sign Pain or weakness Indicates Supraspinatus tendinopathy or tear---

5 DROP ARM TEST Purpose Assess rotator cuff tear Procedure Passively abduct the arm and ask the patient to slowly lower it Positive Sign Arm drops suddenly or cannot be lowered smoothly Indicates Supraspinatus tear/Rotator cuff tear---

III LABRAL LESION TESTS --- **1 ACTIVE COMPRESSION TEST OBRIEN TEST** Purpose Assess SLAP lesion and AC joint pathology Procedure Shoulder flexed to 90 adducted 10 15 Thumb pointing downward Apply downward resistance Repeat with palm facing upward Positive Sign Pain or clicking that decreases with the palm up position Indicates SLAP lesion or AC joint pathology--

2 KIM TEST Purpose Assess posteroinferior labral lesion Procedure Shoulder abducted to 90 Apply axial compression with upward and posterior force while elevating the arm diagonally Positive Sign Posterior shoulder pain Indicates Posteroinferior labral tear---

IV SCAPULAR DYSKINESIA TESTS --- **1 SCAPULAR DYSKINESIS TEST** Purpose Assess abnormal scapular movement Procedure Patient performs repeated shoulder flexion and abduction with light weights while the examiner observes scapular motion Positive Sign Scapular winging or asymmetrical movement Indicates Scapular dyskinesia---

2 SCAPULAR LOAD TEST SCAPULAR ASSISTANCE TEST Purpose Assess the effect of scapular assistance on shoulder movement Procedure Examiner manually assists upward rotation and posterior tilt of the scapula during arm elevation Positive Sign Pain decreases or shoulder movement improves Indicates Scapular dyskinesia---

V MUSCLE PATHOLOGY TESTS --- **1 YERGASONS TEST** Purpose Assess biceps tendon pathology Procedure Patient supinates the forearm against resistance with the elbow flexed to 90 Positive Sign Pain or snapping over the bicipital groove Indicates Biceps tendinitis or transverse humeral ligament injury---

2 SPEEDS TEST Purpose Assess long head of biceps tendon Procedure Shoulder flexed to 90 elbow extended forearm supinated resist downward pressure Positive Sign Pain in the bicipital groove Indicates Biceps tendinitis or SLAP lesion---

3 LIFT OFF TEST Purpose Assess Subscapularis muscle Procedure Place the dorsum of the hand on the lower back and ask the patient to lift the hand away Positive Sign Unable to lift the hand or marked weakness Indicates Subscapularis tear or weakness---

4 TEST FOR SUPRASPINATUS Purpose Assess supraspinatus muscle Procedure Perform the Empty Can Test against resistance Positive Sign Pain or weakness Indicates Supraspinatus pathology---

5 TEST FOR INFRASPINATUS Purpose Assess infraspinatus muscle Procedure Patient externally rotates the shoulder against resistance with the elbow flexed to 90 Positive Sign Pain or weakness Indicates Infraspinatus tear or weakness---

6 HORNBLLOWERS SIGN Purpose Assess Teres Minor function Procedure Shoulder elevated to 90 in the scapular plane with elbow flexed to 90 Ask the patient to maintain external rotation Positive Sign Unable to maintain the position or external rotation Indicates Teres minor tear or severe rotator cuff pathology

5 Special Test: Elbow Joint: Test for ligament instability: Valgus- Varus ligamentary Test etc. Test for Tennis Elbow: Cozens, Mill's, Maudsley's etc. Test for Golfer's Elbow. Test for Neurological symptoms: Pinch Grip Test, Tinel Sign etc. PFDP 5 Elbow Special Tests.pdf ---

I TESTS FOR LATERAL EPICONDYLITIS TENNIS ELBOW --- **1 COZENS TEST** Purpose To assess lateral epicondylitis Tennis Elbow Procedure • Patient makes a fist • Elbow flexed to 90 • Forearm pronated • Wrist extended and radially deviated against resistance Positive Sign Pain over the lateral epicondyle Indicates Lateral epicondylitis ---

2 MILLS TEST Purpose To assess lateral epicondylitis Procedure • Patient sits with the elbow extended • Forearm pronated • Therapist passively flexes the wrist while maintaining elbow extension Positive Sign Pain over the lateral epicondyle

Indicates Lateral epicondylitis --- **3 MAUDSLEYS TEST MIDDLE FINGER TEST** Purpose To assess lateral epicondylitis especially involvement of the Extensor Digitorum Communis EDC Procedure • Elbow extended • Forearm pronated • Patient resists extension of the middle finger Positive Sign Pain over the lateral epicondyle Indicates Lateral epicondylitis --- **II TEST FOR MEDIAL EPICONDYLITIS GOLFERS ELBOW** --- **4 GOLFERS ELBOW TEST MEDIAL EPICONDYLITIS TEST** Purpose To assess medial epicondylitis Golfers Elbow Procedure • Patient sits with the elbow extended • Forearm supinated • Therapist passively extends the wrist and fingers while maintaining elbow extension Positive Sign Pain over the medial epicondyle Indicates Medial Epicondylitis Golfers Elbow --- **III NERVE COMPRESSION TEST** --- **5 TINELS SIGN CUBITAL TUNNEL** Purpose To assess ulnar nerve irritation or compression Procedure • Slightly flex the elbow • Tap gently over the ulnar nerve behind the medial epicondyle Positive Sign Tingling or electric shock sensation along the ring and little fingers Indicates Cubital Tunnel Syndrome Ulnar nerve entrapment --- **IV LIGAMENT STABILITY TESTS** --- **6 VALGUS STRESS TEST** Purpose To assess the Medial Ulnar Collateral Ligament MCL/UCL Procedure • Elbow slightly flexed 20 30 • Stabilize the humerus • Apply a valgus force to the forearm Positive Sign Pain or excessive medial joint opening Indicates Medial collateral ligament injury -- **7 VARUS STRESS TEST** Purpose To assess the Lateral Collateral Ligament LCL Procedure • Elbow slightly flexed 20 30 • Stabilize the humerus • Apply a varus force to the forearm Positive Sign Pain or excessive lateral joint opening Indicates Lateral collateral ligament injury

6 Special Test: Wrist & Hand Complex: Test for Carpal Tunnel Syndrome: Carpal Compression test, Hand Elevation Test, Phalen's, Froment's sign etc. Test for Bone, Ligament, Capsule & joint Instability: Ulnar fovea Test, Ulnar Menisco Triquetral dorsal Glide Test, Watson Test, Bunnel-Littler Test etc. Test for circulation & Swelling: Allen Test, Digital blood flow, Figure of 8 measurement of swelling. Hand volume Test etc. PFDP 6 Wrist Special Tests .pdf --- **WRIST SPECIAL TESTS** --- **I CARPAL TUNNEL SYNDROME TESTS** --- **1 PHALENS TEST** Purpose To assess Carpal Tunnel Syndrome Median nerve compression Procedure • Ask the patient to flex both wrists maximally • Dorsum of both hands are pressed together • Hold for 60 seconds Positive Sign Pain tingling or numbness in the thumb index middle and lateral half of ring finger Indicates Carpal Tunnel Syndrome --- **2 REVERSE PHALENS TEST PRAYER TEST** Purpose To assess Carpal Tunnel Syndrome Procedure • Ask the patient to press both palms together • Maintain maximum wrist extension for 60 seconds Positive Sign Pain or paresthesia in the median nerve distribution Indicates Carpal Tunnel Syndrome --- **3 TINELS SIGN WRIST** Purpose To assess Median nerve irritation Procedure • Tap gently over the median nerve at the carpal tunnel Positive Sign Tingling or electric shock sensation in the median nerve distribution Indicates Carpal Tunnel Syndrome --- **II TENDON PATHOLOGY TEST** --- **4 FINKELSTEINS TEST** Purpose To assess De Quervains Tenosynovitis Procedure • Patient places the thumb inside the fist • Fingers wrap around the thumb • Examiner passively deviates the wrist towards the ulnar side Positive Sign Pain over the radial styloid Indicates De Quervains Tenosynovitis --- **III LIGAMENT STABILITY TEST** --- **5 WATSON SCAPHOID SHIFT TEST** Purpose To assess Scapholunate ligament instability Procedure • Apply pressure over the scaphoid tubercle • Move the wrist from ulnar deviation to radial deviation Positive Sign Pain or palpable clunk Indicates Scapholunate ligament injury --- **IV THUMB TEST** --- **6 GRIND TEST** Purpose To assess 1st Carpometacarpal CMC joint osteoarthritis Procedure • Apply axial compression to the thumb • Simultaneously rotate the thumb metacarpal Positive Sign Pain or crepitus Indicates 1st CMC joint osteoarthritis --- **I TESTS FOR BONE LIGAMENT CAPSULE JOINT INSTABILITY** --- **1 ULNAR FOVEA TEST** Purpose To assess Triangular Fibrocartilage Complex TFCC injury Procedure • Press the examiners thumb into the ulnar fovea between the ulnar styloid and flexor carpi ulnaris tendon Positive Sign Pain or tenderness over the ulnar fovea Indicates TFCC injury or ulnotriquetral ligament injury --- **2 ULNO MENISCO TRIQUETRAL DORSAL GLIDE TEST** Purpose To assess ulnotriquetral ligament and TFCC pathology Procedure • Stabilize the distal ulna • Apply a dorsal glide to the triquetrum while assessing pain and movement Positive Sign Pain or excessive movement Indicates Ulnotriquetral ligament injury or TFCC lesion --- **3 BUNNELL LITTLER TEST** Purpose To differentiate intrinsic muscle tightness from joint capsule tightness Procedure • Flex the PIP joint with the MCP joint extended • Repeat with the MCP joint flexed Positive Sign • PIP flexion improves when MCP is flexed Intrinsic muscle tightness • PIP flexion remains limited Capsular tightness Indicates Intrinsic muscle contracture or capsular restriction

7 Special Test: Lumbo-Sacral region: Test for Lumbar Instability: Passive lumbar extension Test etc. Test for Muscle Dysfunction: 90-90 Straight leg rising, Trendelenburg Test etc. Test for Neurological Dysfunction: SLR, Slump Test etc. Test for Joint Dysfunction: Schober's, Faber- Patrick's, Gaenslen, Gillet, March's Test etc. PFDP 7 lumbosacral special tests.pdf **SPECIAL TESTS LUMBO SACRAL REGION** --- **I TEST FOR LUMBAR INSTABILITY** --- **1 PASSIVE LUMBAR EXTENSION TEST** PURPOSE To assess lumbar spine instability PROCEDURE • Patient lies prone • Examiner lifts both the patients legs together by holding around the ankles • The legs are elevated while the knees remain extended • Observe the patients

symptoms and lumbar response POSITIVE SIGN • Development/reproduction of low back pain • Feeling of heaviness or instability in the lumbar region INDICATES Lumbar segmental instability --- **II TESTS FOR MUSCLE DYSFUNCTION** --- **2 90 90 STRAIGHT LEG RAISING TEST** PURPOSE To assess hamstring muscle tightness/flexibility PROCEDURE • Patient lies supine •

Flex the hip and knee to approximately 90 • Stabilize the thigh • Slowly extend the knee while maintaining hip flexion
POSITIVE SIGN • Inability to achieve full knee extension • Increased resistance/tightness in the posterior thigh **INDICATES** Hamstring tightness/shortness --- **3 TRENDLENBURG TEST** **PURPOSE** To assess hip abductor muscle function especially gluteus medius **PROCEDURE** • Patient stands on one leg • Ask the patient to lift the opposite foot from the floor • Observe the position of the pelvis **POSITIVE SIGN** • Pelvis drops on the side of the lifted/non weight bearing leg **INDICATES** Weakness of the hip abductors of the weight bearing leg particularly gluteus medius --- **III TESTS FOR NEUROLOGICAL DYSFUNCTION** -
 -- **4 STRAIGHT LEG RAISING TEST SLR/LASÈGUES TEST** **PURPOSE** To assess lumbosacral nerve root irritation particularly due to PIVD **PROCEDURE** • Patient lies supine • Keep the knee fully extended • Slowly raise the patients straight leg by flexing the hip • Note the angle and symptoms produced **POSITIVE SIGN** • Reproduction of radicular/sciatic pain typically between about 30 70 of hip flexion **INDICATES** Lumbosacral nerve root irritation/compression commonly associated with PIVD --- **5 SLUMP TEST** **PURPOSE** To assess neural tissue mechanosensitivity particularly lumbosacral nerve root involvement **PROCEDURE** Patient sits with 1 Hands behind back 2 Thoracic/lumbar spine flexed 3 Cervical spine flexed 4 One knee is extended 5 Ankle is dorsiflexed The examiner progressively adds these components while observing symptoms **POSITIVE SIGN** • Reproduction of the patients familiar neural/radicular symptoms • Symptoms change with release of one component particularly cervical extension or ankle movement **INDICATES** Neural tissue/nerve root involvement such as lumbar radiculopathy --- **IV TESTS FOR JOINT DYSFUNCTION** --- **6 SCHOBERS TEST** **PURPOSE** To assess lumbar spine flexion/mobility **PROCEDURE** • Patient stands erect • Mark the lumbosacral region and a point 10 cm above it • Ask the patient to bend forward maximally • Measure the distance between the marks again **POSITIVE SIGN** Insufficient increase in distance during forward flexion **INDICATES** Reduced lumbar flexion/mobility commonly seen in conditions such as ankylosing spondylitis --- **7 FABER / PATRICKS TEST** **FABER** Flexion ABduction External Rotation **PURPOSE** To assess hip and sacroiliac joint pathology **PROCEDURE** • Patient lies supine • Place the tested leg in • Hip flexion • Abduction • External rotation • Place the lateral ankle over the opposite thigh • Stabilize the opposite pelvis • Gently press the flexed knee downward **POSITIVE SIGN** • Pain reproduced in the hip/groin suggests hip pathology • Pain in the sacroiliac region may indicate SI joint pathology • Restricted ROM may also be significant **INDICATES** Hip or sacroiliac joint dysfunction --- **8 GAENSLER TEST** **PURPOSE** To assess sacroiliac joint dysfunction/pathology **PROCEDURE** • Patient lies near the edge of the table • One hip and knee are maximally flexed toward the chest • The opposite leg is allowed to hang off the edge of the table • Apply gentle pressure to produce opposite movements at the pelvis **POSITIVE SIGN** • Reproduction of pain in the sacroiliac region **INDICATES** Sacroiliac joint dysfunction --- **9 GILLETTS TEST** **PURPOSE** To assess sacroiliac joint mobility **PROCEDURE** • Patient stands • Examiner palpates the PSIS and sacrum • Patient flexes one hip and knee lifting the foot from the floor • Observe movement of the PSIS relative to the sacrum **POSITIVE SIGN** • Abnormal or restricted movement of the PSIS compared with the opposite side **INDICATES** Sacroiliac joint hypomobility/dysfunction --- **10 MARCHS TEST** **PURPOSE** To assess sacroiliac joint movement/dysfunction **PROCEDURE** • Patient stands • Examiner palpates the PSIS • Patient marches by lifting one knee at a time • Observe the movement of the PSIS **POSITIVE SIGN** • Abnormal/asymmetrical movement of the PSIS **INDICATES** Sacroiliac joint dysfunction or altered mobility

8 Special Test: Hip Joint: Test For hip Pathology: Hip scour Test, Leg Roll Test, Patrick's etc. Labral Lesion: Anterior Labral tear Test etc. Pediatric Hip Pathology: Barlow's Test, Galeazzi Sign, Ortolani Sign etc. Test For Muscle dysfunction: Hip Lag sign, 90-90 Straight leg rising, Thomas, Nelaton's line, Bryant's triangle, Faber- Gaenslen, Gillet, March's test, Finkelstein Test, Tripod sign, Trendelenburg sign etc. PFDP 8 hip special tests.pdf --- **I TESTS FOR HIP PATHOLOGY** --- **1 HIP SCOUR**

TEST **PURPOSE** To assess for intraarticular hip pathology **PROCEDURE** • Patient lies supine • Flex the hip and knee • Apply a gentle axial load through the femur • Move the hip through flexion/extension and abduction/adduction while maintaining compression **POSITIVE SIGN** • Pain • Catching • Clicking • Crepitus • Restricted/abnormal movement **INDICATES** Possible intraarticular hip pathology such as degenerative or labral pathology --- **2 LEG ROLL TEST** **PURPOSE** To assess for intraarticular hip pathology and restriction of passive hip rotation **PROCEDURE** • Patient lies supine with the leg relaxed • Examiner gently rolls the leg from side to side producing internal and external rotation of the hip • Compare with the opposite side **POSITIVE SIGN** • Pain • Markedly increased or decreased hip rotation • Excessive external rotation compared with the opposite side **INDICATES** Possible hip joint pathology --- **II TEST FOR LABRAL LESION** --- **4 ANTERIOR LABRAL TEAR TEST** **PURPOSE** To assess for an acetabular labral lesion particularly an anterior labral lesion **PROCEDURE** • Patient lies supine • Flex the hip and bring it through the provocative combination of hip movement used to stress the anterior labrum • Reproduce the patients familiar symptoms • Compare with the opposite side **POSITIVE SIGN** • Reproduction of anterior

hip/groin pain • Clicking catching or givingway sensation may occur **INDICATES** Possible acetabular labral lesion --- **III PEDIATRIC HIP PATHOLOGY** --- **5 BARLOWS TEST** **PURPOSE** To determine whether an infants hip is unstable and can be dislocated **PROCEDURE** • Infant lies supine • Flex the hip and knee • Stabilize the pelvis • Gently adduct the hip • Apply gentle posteriorly directed pressure through the femur **POSITIVE SIGN** • A palpable clunk as the femoral head dislocates

INDICATES Dislocatable/unstable hip suggestive of developmental dysplasia of the hip DDH --- **6 GALEAZZI SIGN** PURPOSE To detect apparent femoral shortening particularly due to hip dislocation PROCEDURE • Infant lies supine • Flex both hips and knees to approximately 90 • Place the feet together • Compare the height of both knees POSITIVE SIGN • One knee is lower than the other INDICATES • Developmental hip dislocation/dysplasia • Femoral shortening • Other limb length discrepancy --- **7 ORTOLANI SIGN** PURPOSE To detect a dislocated hip that can be reduced PROCEDURE • Infant lies supine with hips and knees flexed • Stabilize the pelvis • Gently abduct the hip while applying pressure over the greater trochanter in an appropriate direction to reduce the femoral head POSITIVE SIGN • A palpable clunk as the femoral head reduces into the acetabulum INDICATES Reducible hip dislocation DDH Barlow vs Ortolani Barlow OUT dislocates Ortolani IN reduces -- **IV TESTS FOR MUSCLE DYSFUNCTION** --- **8 HIP LAG SIGN** PURPOSE To assess hip extensor muscle function particularly the ability to actively maintain hip extension PROCEDURE • Patient lies prone • Examiner passively extends the hip • Ask the patient to maintain the hip in the extended position after the examiner releases support POSITIVE SIGN • The patient cannot maintain the extended position • The hip falls back toward flexion INDICATES Hip extensor weakness/dysfunction -- **10 THOMAS TEST** PURPOSE To assess hip flexor tightness/hip flexion contracture PROCEDURE • Patient lies supine near the edge of the table • Patient flexes one hip and knee toward the chest to flatten the lumbar spine • Observe the position of the opposite leg POSITIVE SIGN • Opposite thigh rises from the table hip flexor tightness • Knee extends possible rectus femoris tightness • Hip abducts possible TFL/ITB tightness INDICATES Hip flexor muscle tightness/contracture --- **V TESTS FOR HIP DISPLACEMENT** --- **11 NELATONS LINE** PURPOSE To assess the position of the greater trochanter in relation to the pelvis PROCEDURE • Patient lies in the appropriate position with the hip examined • Draw an imaginary line from the ASIS to the ischial tuberosity • Assess the position of the greater trochanter relative to this line POSITIVE FINDING • Greater trochanter lies above Nelatons line INDICATES Possible superior displacement of the femur/hip dislocation --- **12 BRYANTS TRIANGLE** PURPOSE To detect proximal displacement of the greater trochanter and apparent shortening of the femur PROCEDURE • Patient lies supine • From the ASIS draw a perpendicular line to the bed • Draw a line from the greater trochanter to meet this perpendicular • Compare the triangles on both sides POSITIVE FINDING • Shortening of the base of Bryants triangle on the affected side INDICATES Possible hip dislocation or proximal displacement of the femur --- **VI SACROILIAC JOINT TESTS** --- **14 GAENSLER'S TEST** PURPOSE To assess SI joint pathology PROCEDURE • Patient lies near the edge of the table • One hip is flexed toward the chest • The opposite leg hangs over the edge • Apply gentle pressure to produce opposite pelvic movements POSITIVE SIGN • Reproduction of pain around the SI joint INDICATES Sacroiliac joint dysfunction/pathology --- **VII OTHER MUSCLE DYSFUNCTION TESTS** --- **17 TRIPOD SIGN** PURPOSE To assess hamstring tightness PROCEDURE • Patient sits with the legs extended • Examiner observes the patients sitting posture • Patient attempts to sit upright POSITIVE SIGN • Patient leans backward and supports themselves with their hands • Knee flexion may occur because of hamstring tightness INDICATES Hamstring tightness/shortness

9 Special Test: Knee Joint: Tests for Ligament Integrity (Collateral & Cruciate): Valgus Stress, Test, Varus Stress Test, Lachman Test, Anterior Drawer Test, Posterior Drawer Test etc. Tests for Meniscal Injury: McMurray's Test, Apley's Compression & Distraction Test etc. Tests for Joint Effusion & Patellofemoral Dysfunction: Fluctuation Test, Patellar Tap Test, Clarke's Test etc. Other Measurements: Q-Angle PFDP 9 knee special tests.pdf --- **SPECIAL TESTS KNEE JOINT** --- **I TESTS FOR LIGAMENT INTEGRITY** --- **A COLLATERAL LIGAMENTS** --- **1 VALGUS STRESS TEST** PURPOSE To assess the integrity of the medial collateral ligament MCL PROCEDURE • Patient lies supine • Knee is slightly flexed approximately 20 30 to unlock the joint • Stabilize the lateral aspect of the knee • Apply a valgus force to the lower leg while stabilizing the thigh POSITIVE SIGN • Excessive medial joint opening compared with the opposite side • Pain along the medial knee INDICATES MCL injury/insufficiency --- **2 VARUS STRESS TEST** PURPOSE To assess the integrity of the lateral collateral ligament LCL PROCEDURE • Patient lies supine • Knee is slightly flexed approximately 20 30 • Stabilize the medial aspect of the knee • Apply a varus force to the lower leg POSITIVE SIGN • Excessive lateral joint opening • Pain along the lateral knee INDICATES LCL injury/insufficiency --- **B CRUCIATE LIGAMENTS** --- **3 LACHMAN TEST** PURPOSE To assess the integrity of the anterior cruciate ligament ACL PROCEDURE • Patient lies supine • Knee flexed approximately 20 30 • Stabilize the femur with one hand • Hold the proximal tibia with the other hand • Pull the tibia anteriorly POSITIVE SIGN • Excessive anterior translation of tibia • Soft/absent firm end feel compared with the normal side INDICATES ACL injury/tear Very important Lachman is generally more sensitive for ACL injury than the anterior drawer test particularly in the acute knee --- **4 ANTERIOR DRAWER TEST** PURPOSE To assess the ACL PROCEDURE • Patient lies supine • Hip flexed and knee flexed to 90 • Stabilize the foot •

Grasp the proximal tibia with both hands • Pull the tibia anteriorly POSITIVE SIGN • Excessive anterior movement of the tibia compared with the opposite side INDICATES ACL injury/insufficiency --- **5 POSTERIOR DRAWER TEST** PURPOSE To assess the integrity of the posterior cruciate ligament PCL PROCEDURE • Patient lies supine • Knee flexed to approximately 90 • Stabilize the foot • Place both hands around the proximal tibia • Push the tibia posteriorly POSITIVE SIGN • Excessive posterior translation of the tibia INDICATES PCL injury/insufficiency --- **II TESTS FOR MENISCAL INJURY** --- **6 MCMURRAY'S**

TEST PURPOSE To assess for medial and lateral meniscal lesions **PROCEDURE** • Patient lies supine • Fully flex the knee • For medial meniscus externally rotate the tibia and apply valgus stress while extending the knee • For lateral meniscus internally rotate the tibia and apply varus stress while extending the knee • Palpate the joint line **POSITIVE SIGN** • Pain • Palpable/audible click or clunk at the joint line **INDICATES** Possible meniscal tear --- **7 APLEYS COMPRESSION DISTRACTION TEST PURPOSE** To differentiate meniscal pathology from ligament pathology **PROCEDURE** Compression • Patient lies prone • Knee flexed to 90 • Stabilize the thigh • Apply downward compression through the heel • Rotate the tibia internally and externally Distraction • Maintain the same position • Apply traction to the tibia while rotating it **POSITIVE SIGN** Compression positive • Pain reproduced with compression/rotation Suggests meniscal pathology Distraction positive • Pain reproduced with traction/rotation Suggests ligament/capsular pathology --- **III JOINT EFFUSION PATELLOFEMORAL DYSFUNCTION** --- **8 FLUCTUATION TEST PURPOSE** To detect knee joint effusion particularly larger effusions **PROCEDURE** • Patient lies supine with the knee relaxed/extended • Examiner compresses the fluid from one side of the knee toward the opposite side • Observe/palpate for a fluid wave **POSITIVE SIGN** • A palpable/visible fluid wave or fluctuation **INDICATES** Knee joint effusion --- **9 PATELLAR TAP TEST PURPOSE** To detect knee joint effusion **PROCEDURE** • Patient lies supine with knee extended • Gently press/compress the suprapatellar region to move fluid beneath the patella • Tap the patella downward against the femur **POSITIVE SIGN** • Patella rebounds/floats and taps the femur producing a palpable tap **INDICATES** Significant knee joint effusion --- **10 CLARKES TEST / PATELLAR GRIND TEST PURPOSE** To assess patellofemoral dysfunction particularly chondromalacia/patellofemoral pain **PROCEDURE** • Patient lies supine with the knee extended • Examiner places the hand proximal to the patella • Apply gentle downward pressure to the patella • Ask the patient to contract the quadriceps **POSITIVE SIGN** • Pain behind/around the patella • Inability to maintain quadriceps contraction because of pain **INDICATES** Possible patellofemoral dysfunction/chondral pathology --- **IV OTHER MEASUREMENT** --- **11 Q ANGLE PURPOSE** To assess patellofemoral alignment and the lateral pull on the patella **MEASUREMENT** Q angle is formed by the intersection of • Line 1 ASIS centre of patella • Line 2 centre of patella tibial tubercle **PROCEDURE** • Patient positioned with the knee in the appropriate standardized position • Identify and mark • ASIS • Centre of patella • Tibial tubercle • Use a goniometer to measure the angle between the two lines **POSITIVE/ABNORMAL FINDING** An increased Q angle may indicate increased lateral patellar alignment/pull and may be associated with patellofemoral dysfunction **INDICATES** • Patellofemoral malalignment • Predisposition to lateral patellar tracking/instability

10 Special Test: Ankle & Foot Complex: Tests for Ligament Stability: Anterior Drawer Test, Talar Tilt Test etc. Tests for Vascular Conditions (e.g., DVT): Homan's Test, Moses Test etc. PFDP 10 ankle foot special tests.pdf --- **I TESTS FOR LIGAMENT STABILITY** --- **1 ANTERIOR DRAWER TEST ANKLE PURPOSE** To assess the integrity of the anterior talofibular ligament ATFL and lateral ankle stability **PROCEDURE** • Patient sits or lies supine • Keep the ankle in a neutral or slight plantar flexed position • Stabilize the distal tibia with one hand • Grasp the heel/talus with the other hand • Draw the foot/talus anteriorly **POSITIVE SIGN** • Excessive anterior translation compared with the opposite ankle • Soft/absent end feel • Pain may be reproduced **INDICATES** ATFL injury/lateral ankle instability --- **2 TALAR TILT TEST PURPOSE** To assess the integrity of the lateral ankle ligaments particularly the calcaneofibular ligament CFL **PROCEDURE** • Patient sits or lies supine • Stabilize the distal leg • Grasp the calcaneus/heel • Tilt the calcaneus into inversion **POSITIVE SIGN** • Excessive inversion compared with the opposite side • Pain or a soft end feel **INDICATES** CFL injury/lateral ankle instability --- **II TESTS FOR VASCULAR CONDITIONS DVT** --- **3 HOMANS TEST PURPOSE** Traditionally used to assess for deep vein thrombosis DVT of the lower limb **PROCEDURE** • Patient lies supine • Knee is extended • Examiner supports the leg • Passively dorsiflex the ankle **POSITIVE SIGN** • Pain in the calf during passive ankle dorsiflexion **INDICATES** Traditionally considered suggestive of DVT --- **4 MOSES TEST PURPOSE** Traditionally used to assess for DVT **PROCEDURE** • Patient lies supine with the leg relaxed • Compress the calf anteroposteriorly • Then compress the calf laterally **POSITIVE SIGN** • Pain produced by anteroposterior compression but not by lateral compression **INDICATES** Traditionally considered suggestive of DVT

SPECIAL TESTS --- **I TESTS FOR CIRCULATION and SWELLING** **1 ALLEN TEST Purpose** To assess the patency of the radial and ulnar arteries **Procedure** • • • • Patient clenches the fist Compress both radial and ulnar arteries Patient opens the hand Release one artery at a time and observe colour return **Positive Sign** Delayed or absent return of normal hand colour **Indicates** Arterial insufficiency or vascular occlusion **2 DIGITAL BLOOD FLOW TEST Purpose** To assess circulation to the fingers **Procedure** • • Compress the fingertip until it blanches Release the pressure **Positive Sign** Delayed capillary refill greater than 2 seconds **Indicates** Reduced peripheral circulation **3 FIGURE OF 8 MEASUREMENT Purpose** To assess hand and wrist swelling edema **Procedure** • Use a measuring tape in a standardized figure of 8 pattern around the wrist and hand Record the measurement **1** • **Positive Finding** Increased measurement compared with the opposite side **Indicates** Hand or wrist edema **4 HAND VOLUME TEST Purpose** To assess hand edema **Procedure** • • Immerse the hand into a volumeter filled with water Measure the volume of displaced water **Positive Finding** Increased displaced volume **Indicates** Hand swelling/edema --- **II TESTS FOR THORACIC OUTLET SYNDROME** **1 ADSONS TEST Purpose** To assess

Thoracic Outlet Syndrome TOS **Procedure** • • Palpate the radial pulse Patient extends the neck rotates the head toward the tested side and takes a deep breath **Positive Sign** Reduced/absent radial pulse or reproduction of symptoms **Indicates** Thoracic Outlet Syndrome **2 ROOS STRESS TEST EAST Purpose** To assess Thoracic Outlet Syndrome **Procedure** Patient abducts shoulders to 90 degrees externally rotates them and flexes elbows to 90 degrees 2 • Patient repeatedly opens and closes the hands for 3 minutes **Positive Sign** Pain numbness tingling heaviness or inability to continue **Indicates** Thoracic Outlet Syndrome